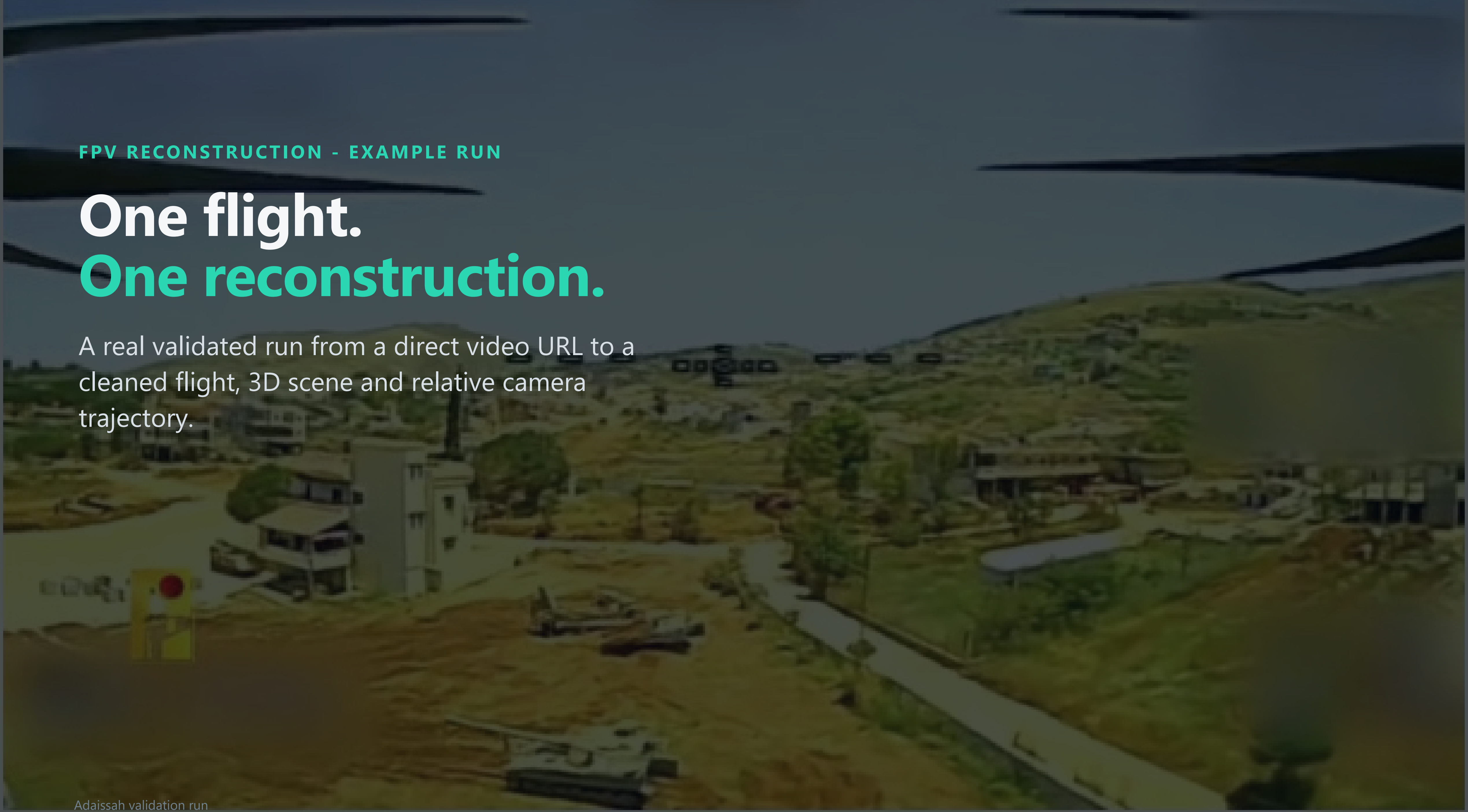


FPV RECONSTRUCTION - EXAMPLE RUN

One flight.
One reconstruction.

A real validated run from a direct video URL to a cleaned flight, 3D scene and relative camera trajectory.

STEP 1 - INPUT

The workflow begins with one direct video URL


```
https://d2fioemadmrru3.cloudfront.net/videos/
2026-06-06_sholef_howitzer_adaissah.mp4
```

53.04s
source duration

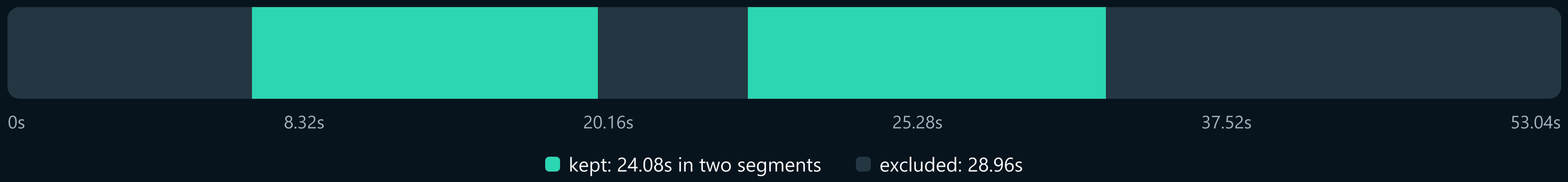
25
frames per second

848×478
source resolution

No manual trimming or frame picking is required before the run.

STEP 2 - VIDEO PREPARATION

Only continuous, useful flight evidence is retained



Detect

Sustained near-identical frames and replay-like repetition are measured across time.

Keep

Segments with continuous forward flight and usable visual change are retained.

Do not invent

Removed motion stays missing; the notebook never fills gaps with duplicated frames.

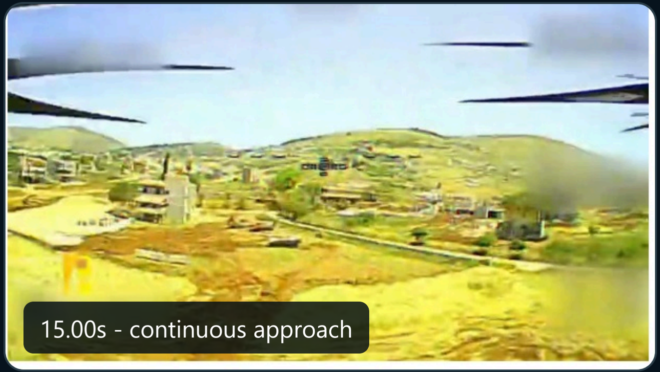
> The decision is not based on one still frame. It requires a sequence of evidence, which prevents a normal brief pause from being treated as a freeze.

Source intervals: 8.32-20.16s and 25.28-37.52s

STEP 3 - CLEANED FLIGHT

The retained video follows the actual approach


8.32s - segment 1 starts


15.00s - continuous approach


26.44s - segment 2


35.96s - close approach

24.08s

usable flight video

The scene remains visually connected from the wider terrain view to the final approach.

PLAY CLEANED VIDEO

Real retained frames - no reconstructed imagery

STEP 4 - FRAME SELECTION

The model receives a compact set of overlapping views

99
candidate frames decoded

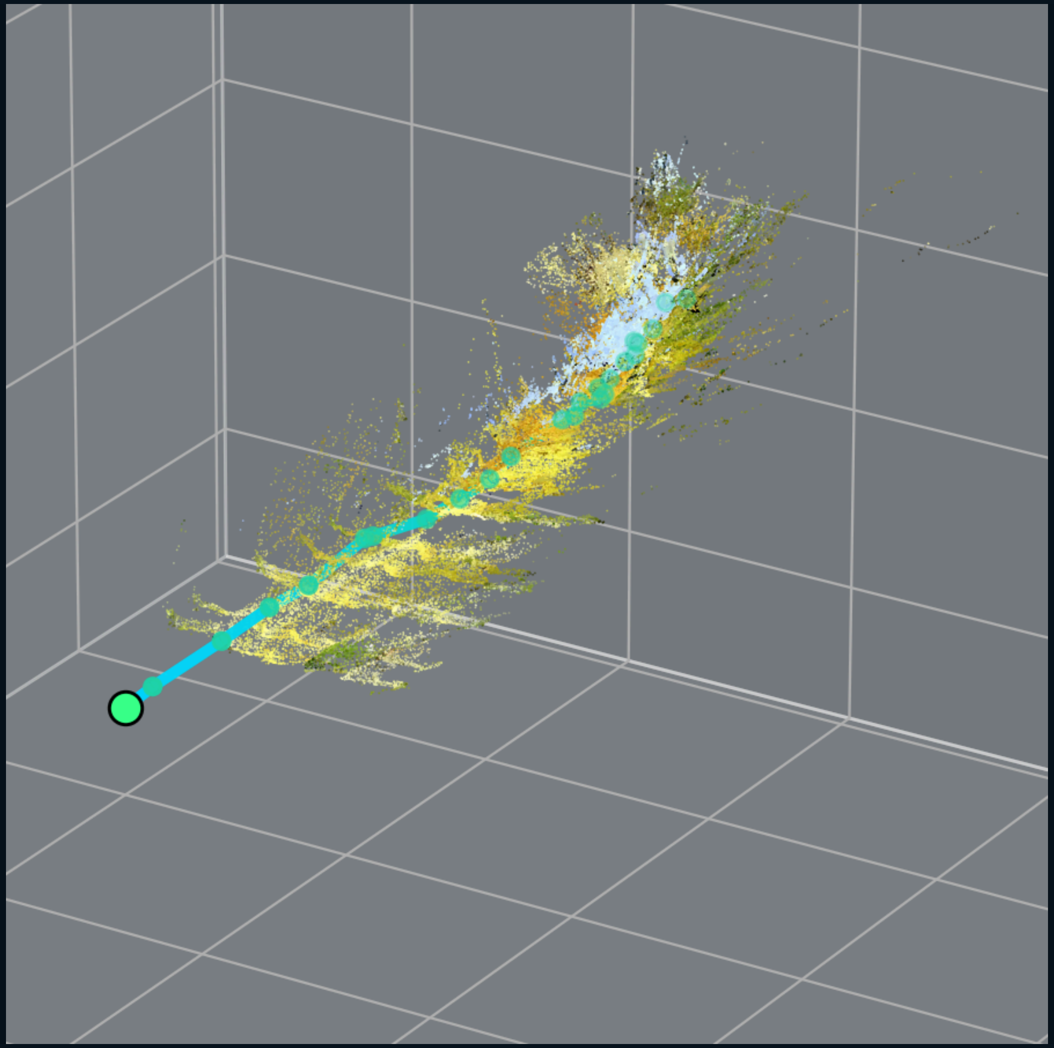
25
views selected for VGGT-Omega

- ✓ Preserves the beginning, middle and end of each retained segment.
- ✓ Favors visual change while maintaining overlap between neighboring views.
- ✓ Avoids spending GPU memory on many nearly identical frames.

More frames are not automatically better. The default balances scene coverage, overlap and T4 memory.

STEP 5 - VGGT-OMEGA RECONSTRUCTION

The selected views become one relative 3D scene



750,000
exported colored 3D points

1
connected geometry component

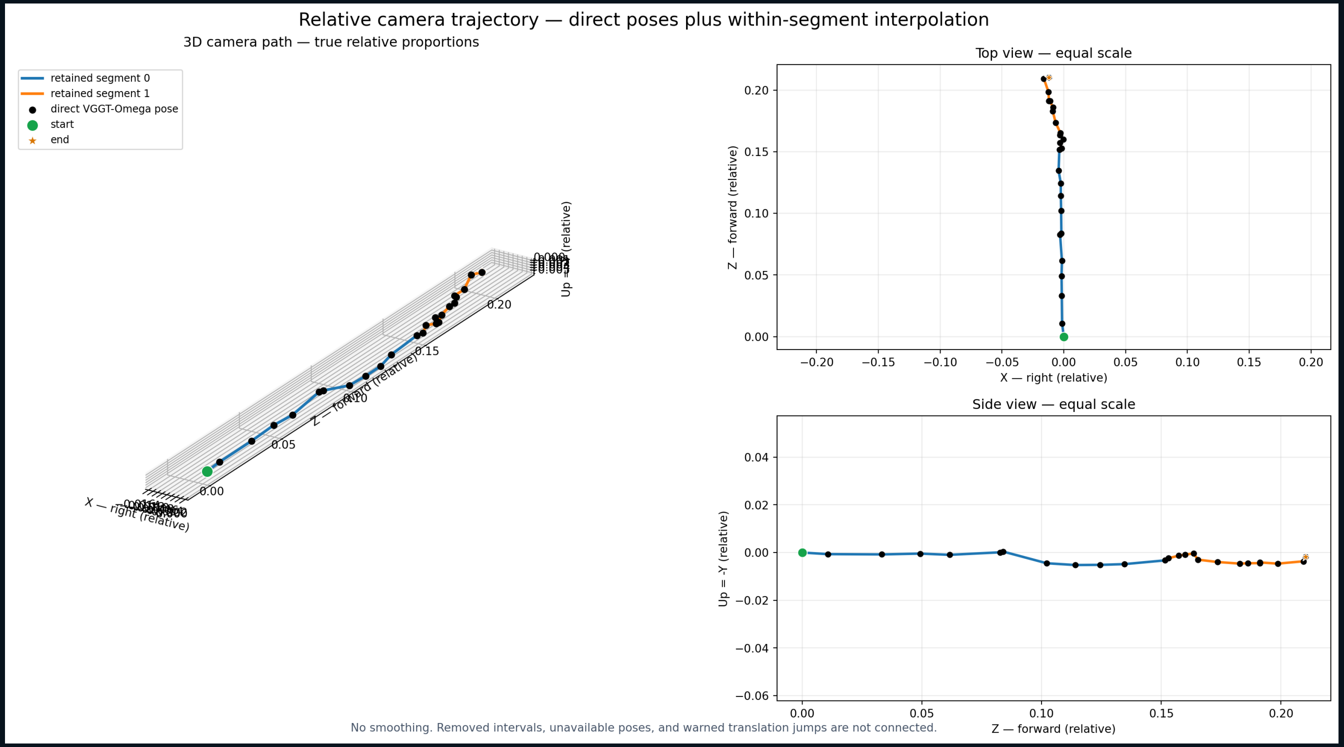
25 / 25
trusted model poses

The colored points represent the reconstructed scene. The turquoise markers show the estimated camera path.

OPEN 3D GLB

STEP 6 - TRAJECTORY AND DELIVERABLES

The final path is shown at true scale



OK 602 retained per-frame rows

OK 25 direct VGGT pose anchors

OK 547 interpolations only inside supported segments

OK 30 rows intentionally unavailable across gaps

IMPORTANT LIMIT: X, Y, and Z are relative model coordinates. They are not metres, GPS, or absolute altitude.

GLB | reconstruction PNG | XYZ CSV | full ZIP

No smoothing. Removed intervals, unavailable poses, and warned jumps are not connected.